

Structural Consequence Model

A Formal Note on Persistence Under Entropy

Abstract. We introduce a minimal structural model describing consequence in systems operating under persistent entropy. The model applies to any domain in which retained signal competes with decay and consequential outcomes accumulate over time.

1 Definitions

Let $t \in \mathbb{R}_{>0}$ denote time. Define

$$X(t) := \text{retained signal at time } t \quad (1)$$

$$C(t) := \text{entropy (churn) at time } t \quad (2)$$

$$S(t) := \text{structural consequence at time } t \quad (3)$$

Assume entropy is strictly positive:

$$\forall t \in \mathbb{R}_{>0}, \quad C(t) > 0 \quad (4)$$

2 Instantaneous Structural Strength

$$S := \frac{X}{C} \quad (5)$$

Consequence increases with signal and decreases with entropy.

3 Cumulative Consequence

$$S(t) = t \cdot \frac{X(t)}{C(t)} \quad (6)$$

If $X(t)$ and $C(t)$ remain approximately stable, consequence increases linearly with time.

4 Derived Rate Form

$$\frac{S(t)}{t} = \frac{X(t)}{C(t)} \quad (7)$$

Average consequence per unit time equals structural signal relative to entropy. Time integrates structural strength; it does not independently generate consequence.

5 Conclusion

The model applies to any domain in which consequential outcomes are expected. Whenever a system operates under entropy, durable consequence emerges from the ratio of structured signal to decay, accumulated across time. Sustained coherence relative to entropy determines the magnitude of enduring impact.